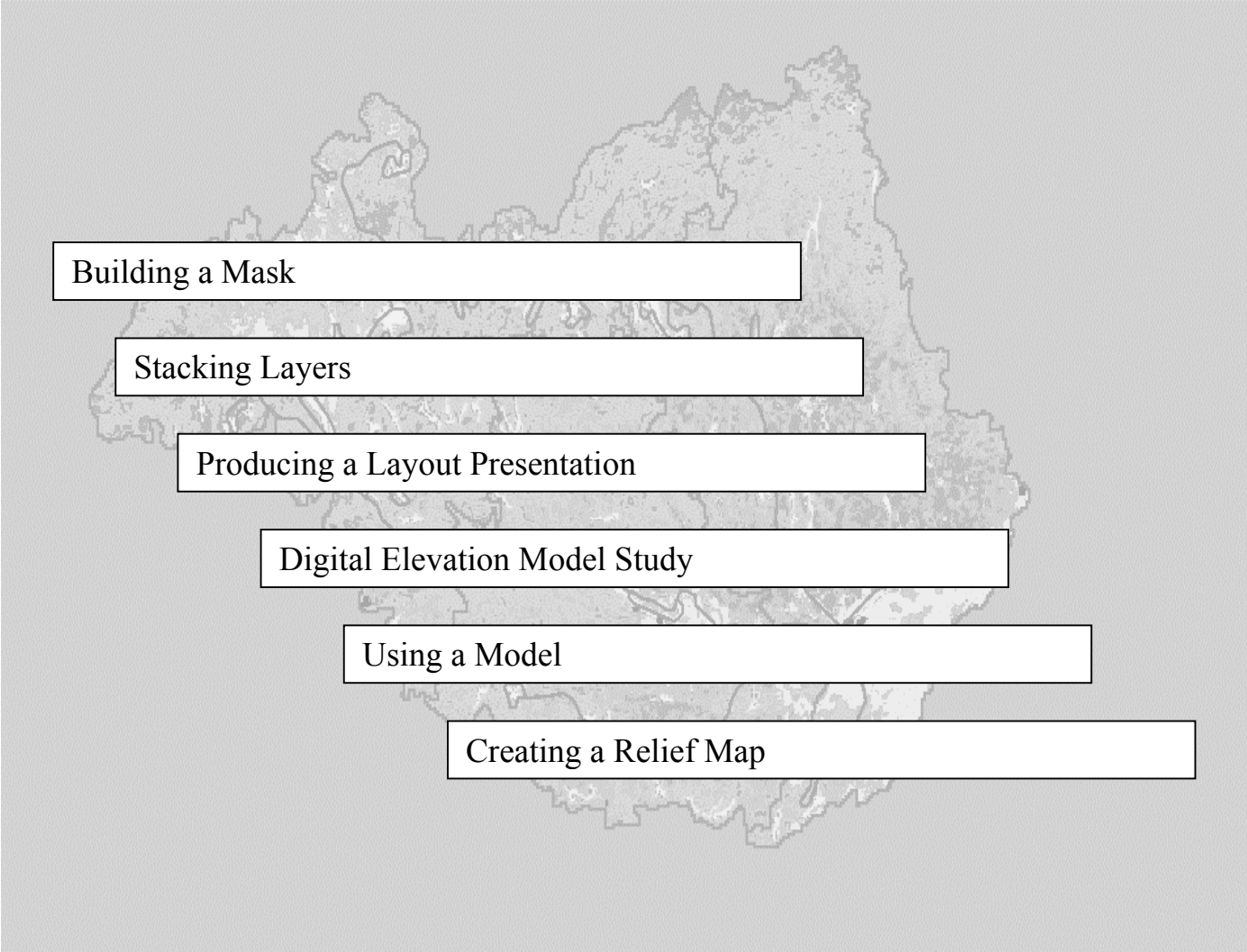


Water Resources and Pollution in Canada



Building a Mask

Stacking Layers

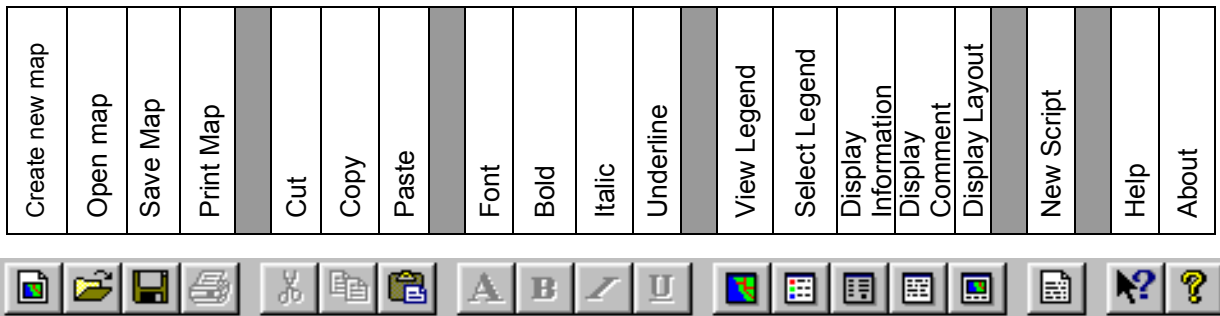
Producing a Layout Presentation

Digital Elevation Model Study

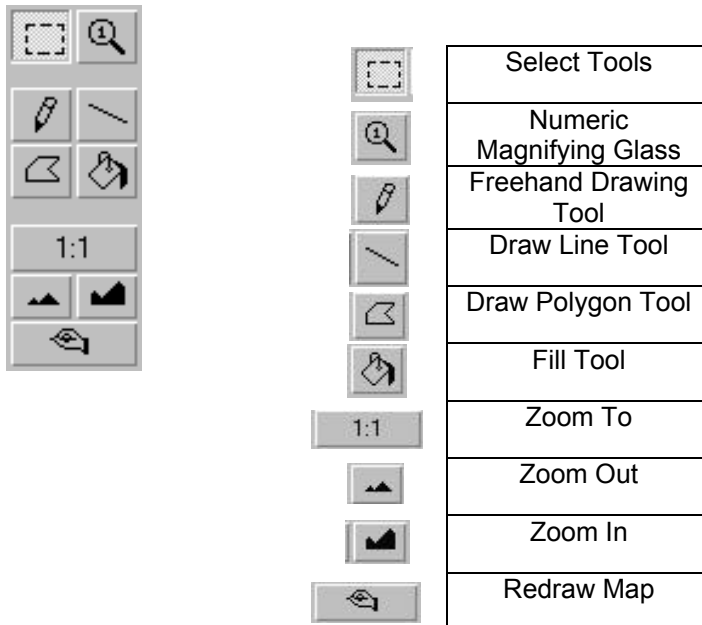
Using a Model

Creating a Relief Map

MFworks Icons and Tool Bars



Tool Bar



For more information check: <http://www.thinkspace.com>

STARTING MFworks

Before you begin, the lesson data folder must be **copied** into a drive in which the student has writing and editing capabilities. This will probably be the drive containing your own personal account on the school's system or it may be the C: **drive** of the machine you are currently working on. If the latter is the case you will have to work on the same machine each day until this assignment is complete. Each school set-up is different so work with your teacher to prepare your data folder.

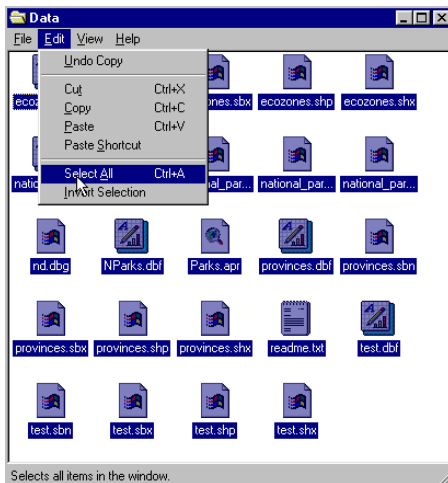
! *(The data for the lesson must first be downloaded from this Web site and placed on the school computer or network. Ask your teacher where to find the data folder that has been downloaded).*

To do this, use the file manager program on your computer. For windows based systems this will be **My Computer / Windows Explorer /**.

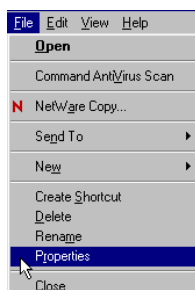
Once the data folder has been **copied** to the required destination the files must be made usable by removing the **read only** attribute. For windows based systems follow these instructions:

! *(The data downloaded from the Web Site is set as **read only** so that it cannot be changed or altered by accident.)*

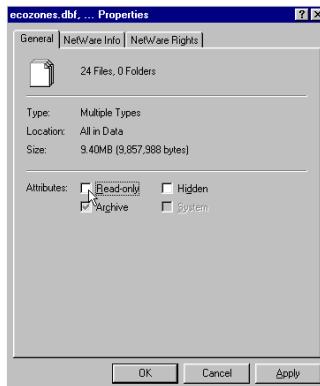
1. Open your new data folder.
2. Choose **Edit > Select All** (they will all appear Blue).



3. From the menu choose:
File > Properties



4. Remove the check mark in the box beside the label ***read only***.



5. Click **Apply**, then **OK**.

The data in your new data folder can now be opened, edited and saved using MF Works to carry out the following project.

! Denotes an observation.

? Denotes a question that may be answered on the Student Activity Sheet.

Begin the lesson by starting up the **MFworks** program.

1. From your system Start menu, choose **Programs > MFworks > MFworks**.

The MFworks program will launch. If necessary, click the window maximize button so that the MFworks window fills the screen.

! *When the program starts, it automatically opens a window titled "**Untitled Project**." MFworks uses **projects** to group together and organize maps that are used together. A project has been created for you to use with this activity, so you do not need the default project.*

2. Choose **File > Close** to close the Untitled Project window.
3. Choose **File > Open Project** to bring up the Open dialog box.
4. Make sure that the folder named in the "**Look in**" field is the data folder just created.
5. Select the file name **Water_Basin.mfp** and click the Open button to open that project.

! *The **Water_Basin.mfp** project lists nine maps that you will use in this activity. You will also create several others using MFworks operations. As they are created, these new maps will automatically be listed in the Project window. In order to keep the workspace usable it is important to close or minimize any maps not in use leaving only one map open on the screen to work on. If you close a map ignore the "Save Changes" message where appropriate.*



This lesson looks at the Ottawa River Watershed using MFworks. It deals with building masks, created slope and shaded relief maps from DEM, composite overlays and data analysis.

BUILDING A MASK

We will begin by building a watershed mask for the Ottawa River drainage area. A mask limits the viewing area of a map by covering all cells that are not related to the defined feature area such as a drainage basin.

1. Double Click **basinlines**. In the project window. This will open the basinlines map and legend windows.
2. Click and Drag the edge or corner of the “**basinlines**” map window to enlarge the Map window to fill your screen.
3. Click on **Edit > Select All**.
4. Click on **Edit > Copy**.
5. Click on **Edit > Paste To New Map**. This will create a new Map “**MapLayer1**” which is a copy of the **basinlines** map.
6. Close the original **basinlines** map window.
7. With the New Map *MapLayer1* open and maximized.

8. Click the **Numeric Magnifying Glass**  button. Move the cursor over the map.

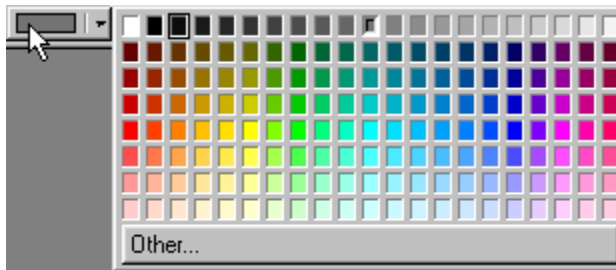
! Check the map to see that all cells are void except the boundary lines that are 1. The Ottawa River drainage area is found on the inside of this boundary line.

Now we want to class the area surrounding the Ottawa River drainage area to 2.

9. Click on the **Fill Area**  button.
10. In the **Paint Value** window at the bottom left of the map enter 2. This will assign the value 2 to the **Fill Area** button.



11. Open the **Paint Colour** list to the left of the window.
12. Select a **medium grey** shade.



13. Move the **Fill Area** cursor to an area outside the watershed.
14. Hold down the **Shift Key** and click the left mouse button.
15. When the area is filled, use the scroll bars to move the map in the map window. You will see new areas of the map that have not been filled yet. Repeat the fill procedure until all the area outside the Ottawa River drainage area is coded 2.

STACKING LAYERS

Now we will add the Ottawa River watershed mask that we just created to the AVHRR land cover map.

! AVHRR stands for Advanced Very High Resolution Radiometer. This is an instrument on a satellite that takes the images of the earth's surface. From these images, the data recorded by the instrument is classified into groups that represent land cover categories.

1. Open the **Water_Basin.mfp project window**.
2. Double click **landcover**. To open the landcover map.
3. Click and Drag the edge or corner of the "**landcover**" map window to enlarge the Map.
4. On the **landcover** map note that built-up areas are black and water is white.

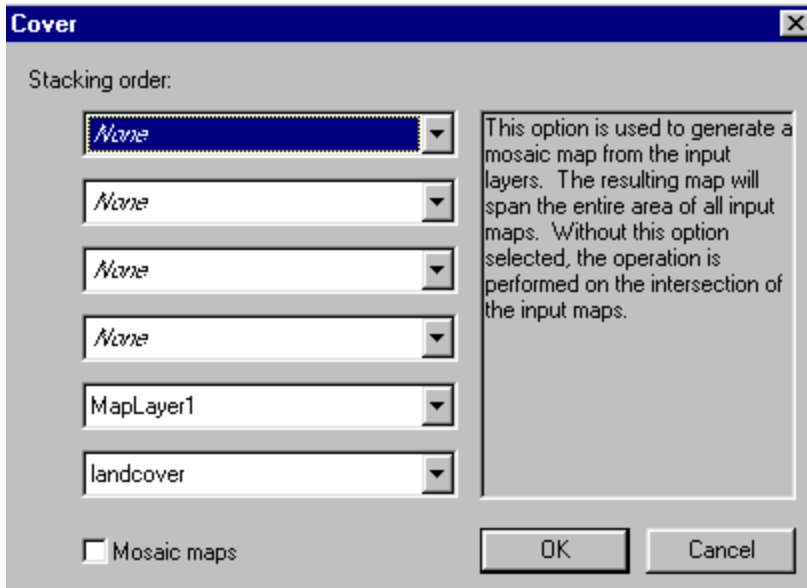
? Can you find the cities of Montreal and Ottawa? Describe where they are.

? Can you find the Ottawa River and the St. Lawrence River? Describe these shapes and the visual appearance of these two features?

5. Click the **Legend**  button. Note the colours and values for *agriculture (Yellow)* and *forest (Greens & Browns)*.
6. Click the **Map**  button.

? Can you describe the distribution of agriculture in the image?

7. Click on **Operation > Cover**.
8. Open the **Map List** at the bottom of the stacking order.
9. Select **landcover**.
10. Open the second **Map List** from the bottom.
11. Select map **layer1**, which you created earlier.
12. Click **O.K.**

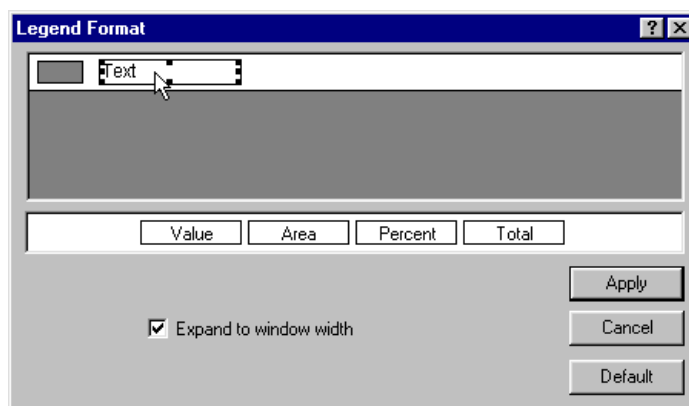


- ! Describe the new map MapLayer2.
- ? What can you say about the landcover within the Ottawa River watershed?
- ! Noting that the red line separates sub-basins, find the Lower (containing Ottawa) basin, Central basin, and Upper basin.
- ! Describe the landcover in each.
- ? If agricultural runoff is a major source of pollution, which sub-basin is at greatest risk? Why?
- ! If acidic soils have low buffering capability, that is to say, they cannot neutralize acidic rain as easily as basic soils, then they are more susceptible to the effect of acid rain. We know that coniferous forests prefer acidic rock and soil conditions. Looking at the landcover map, find the areas of acidic rock and soil by their association with coniferous forests.
- ? Which sub-basin is most affected by acid rain?

PRODUCE A LAYOUT PRESENTATION

We will now prepare a layout presentation of the Landcover map that can be printed.

1. Click **Layout**  button.
2. Centre the map on the page.
Edit the legend as follows:
3. Click **Legend** button.
4. Click **Legend > Format**.
5. Click and drag **Value** and **Area** from the top row down to the bottom row.
6. Click and drag **Text** over next to **Symbol Colour** box.
7. **Apply**.



8. Click **Layout**  button.
9. Click **Legend Style**  tool button.
10. Add legend.

11. Click **add text**  tool button.
12. Add title.

DIGITAL ELEVATION MODEL STUDY

Let's study the digital elevation model of the Ottawa River watershed. This map will show us the relative elevations of the land surrounding the Ottawa River watershed cell by cell. We will see how the water basin collects the precipitation and funnels the water down to the river.

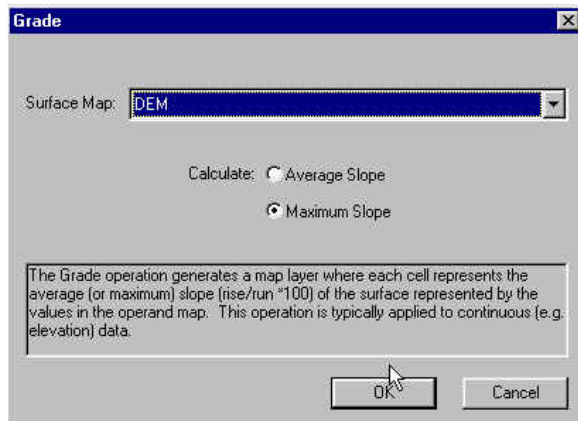
1. Click MENU item **window > Water_Basin**.
2. Double click **DEM. (Digital Elevation Model)**.
3. Use the **Numeric Magnifying Glass**  to investigate the values in the DEM (digital elevation model). The values measure the mean elevation for each cell in meters above mean sea level.

- ? Can you identify some of the major topographic features?
- ? Where is the highest point?
- ? Where is the lowest?
- ? Can you trace some of the runoff patterns? Describe them here.

CREATING A SLOPE MAP

A slope map shows us the relative angle or slope between the high points and the low points on the map. The greater the angle of the slope the lighter is the image.

1. Click MENU item **operation > Grade**.
2. Open **Surface Map** list.
3. Select **DEM**.
4. Click **Maximum Slope** button.
5. Click **O.K.**



- ? Can you see a pattern between the runoff and the slope?

? If steeper slopes are good for ski hills, where would you expect to find the ski hills in this watershed?

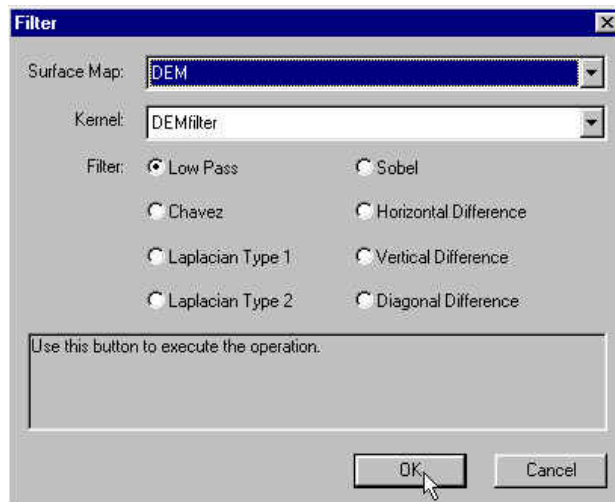
CREATING A SHADED RELIEF MAP

A shaded relief map shows the relief or elevation pattern of an area by means of creating shadows. It simulates the shadows that would be observed on the ground if the sun were shining. The higher a hill or mountain, the longer is the shadow.

1. Click **Window > Water_Basin**.
2. Double click **DEM filter**.
3. Use the **Numeric Magnifying Glass** to study the values in this map to be used as a kernel in a filtering routine.

- ? What is the maximum value?
- ? What is the minimum value?
- ? What is the pattern of values?
- ? What is the size of the *kernel*?

3. Click **Operation > Filter**.
4. Open **Surface Map** list.
5. Select **DEM**.
6. Open **Kernel** list.
7. Select **DEM filter**.
8. Click **O.K.**



Study the shaded relief map.

- ? What direction is the illumination coming from?
- ? What happens if we rotate the filter by 180 degrees?

Repeat the process above but try using the DEMfilter2 as the kernel.

- ? What changes do you see?

PRODUCE A SHADED RELIEF MAP USING A LANDCOVER MAP

1. Click **New Script**  button.
2. Enter the following:

shadedlandcover = relief + *landcover*;

3. Click MENU item **Script. > Check Script.**
4. If there are no errors, click **O.K.**
5. Click MENU item **Script > Execute Script.**

- ? Can you see a relationship between topography and agriculture?

6. Click MENU item **Operation > Cover.**
7. Select **Shadedlandcover** for bottom of stack.
8. Select **MapLayer1** for next layer in the stack.
9. Click **O.K.**

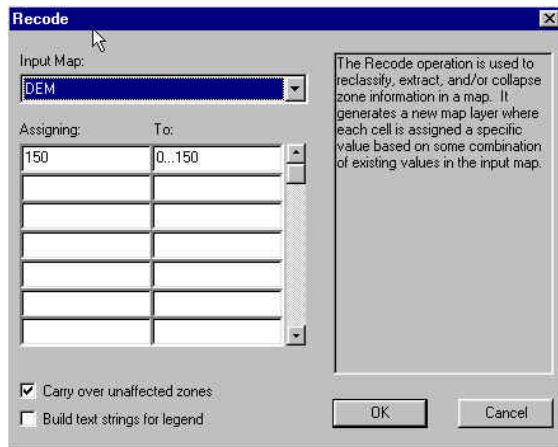
- ? What can you see in the shaded landcover map of the Ottawa River watershed?

USING A MODEL

If global warming caused a 150 metres rise in sea level due to ice cap melting, what part of the Ottawa River watershed would be flooded by seawater? Let's find out.

1. Click MENU item **Operation > Recode.**
2. Open **Input Mask** list.
3. Select **DEM.**
4. Enter **150** in the **Assigned** field.
5. Enter **0...150** in the **To** field.

We now have recoded all the cells with values of 150 metres or less to a single value of 150 metres.



6. Check **Carry Over Unaffected Zones. O.K.**
7. Click **Legend**  button for the new map layer.
8. Click **Symbol Colour** box for 150.



9. Select a **blue** colour.
10. Click **Map**  button.
11. Click MENU item **Operations > Cover**.
12. Select the **New Map Layer** for the bottom of the stack.
13. Select **Mask** for the next layer in the stack.

! Describe the area that would be covered with seawater.